

LABWORK 1: Cinema Ticket Calculator

Write a program that calculates the total cost of a cinema ticket, including discounts, VAT, and service fees, based on information received from the user.

Function 1 (15 pts):

- **Name of the function:** `calculate_subtotal`
- **Return type:** `float`
- **Input parameters:** 2 numbers (`float ticket_price`, `int quantity`)
- **Description:** Returns the subtotal by multiplying the ticket price by the quantity. (Price * Quantity)

Function 2 (15 pts):

- **Name of the function:** `apply_discount`
- **Return type:** `float`
- **Input parameters:** 2 numbers (`float subtotal`, `float discount_percentage`)
- **Description:** Calculates and returns the discount **amount** to be applied to the subtotal. (Subtotal * discount percentage/100)

Function 3 (15 pts):

- **Name of the function:** `calculate_vat`
- **Return type:** `float`
- **Input parameters:** 1 number (`float subtotal`)
- **Description:** Calculates and returns the VAT amount based on the subtotal. (Assume a fixed VAT rate of 20%). (Subtotal * 0.20)

Function 4 (15 pts):

- **Name of the function:** `calculate_service_fee`
- **Return type:** `float`
- **Input parameters:** 1 number (`int quantity`)
- **Description:** Calculates and returns the total service fee. Assume a fixed service fee of 5.0 currency units per ticket. (Quantity * 5.0)

Function 5 (15 pts):

- **Name of the function:** `calculate_total_bill`
 - **Return type:** `float`
 - **Input parameters:** 4 numbers (`float subtotal`, `float discount_amount`, `float vat_amount`, `float service_fee`)
 - **Description:** Calculates and returns the final invoice amount. (Formula: Subtotal - Discount Amount + VAT Amount + Service Fee)
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Main Function (25 pts):

- **Name of the function:** `main`
 - **Return type:** `int`
 - **Input parameters:** none
 - **Description:**
 1. It takes the ticket price, quantity, and discount percentage from the user.
 2. It calculates the subtotal with `calculate_subtotal`.
 3. It finds the discount amount with `apply_discount`.
 4. It calculates the VAT amount with `calculate_vat`.
 5. It calculates the service fee with `calculate_service_fee`.
 6. It calculates the total bill with `calculate_total_bill`.
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Restrictions

- Loops, arrays, and conditional structures (if/else, switch) are strictly forbidden.
- If you are not sure if something is free to use or not, please ask your assistant.
- You have to do your job using the functions given to you. If you complete the labwork without using functions, your work will not be graded.



```
1- /*****
2-
3- Welcome to GDB Online.
4- GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,
5- C#, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, Prolog.
6- Code, Compile, Run and Debug online from anywhere in world.
7-
8- *****/
9- #include <stdio.h>
10-
11- int main()
12- {
13-     printf("Hello World");
14-
15-     return 0;
16- }
17-
```

Submission

- Submit your C file with the format "name_surname.c" (use your name and surname).
- Do not submit a Word document, text file, or executable (a.out).